Financial Time Series
46-929
Homework #1

Due: Tuesday, March 30, 2021, 1:30pm.

1. **Practice in creating time series plots**
   Imitating the graphs presented in class, pick any two of the four stocks: Apple (aapl), Pfizer (pfe), Gamestop (gme) and Bitcoin (btc-usd) and create plots for the log-returns, a q-q plot for the log-returns and an acf plot for the log returns. Do your plots give evidence for or against the stylized facts of Rama Cont?

2. **Behavior of absolute returns**
   One of the stylized principles of Rama Cont is the slow decay of absolute returns: "the autocorrelation funciton of absolute returns decays slowly as a function of the time lag, roughly as a power law with an exponent $\beta \in [0.2, 0.4]$." Explore this by taking a few stocks of your choice and seeing if they follow this stylized fact.

3. Let $\{\epsilon_t,\ -\infty < t < \infty\}$ be a sequence of independent random variables with common mean 0 and common variance σ^2. For each of the three processes $\{X_t,\ -\infty < t < \infty\}$ below, which are weakly stationary processes? For those that are, find the mean and covariance function.

   a) $X_t = a_0 + a_1\epsilon_t + a_2\epsilon_{t-1}$ for constants $a_i,\ i = 1, 2, 3$.
   b) $X_t = a_0 + a_1 f(\epsilon_t)$ for constants a_0, a_1 and some function f.
   c) $X_t = \Pi_{i=0}^{k}\epsilon_{t-i},\ k \geq 1$..

4. Let $\{\epsilon_t,\ -\infty < t < \infty\}$ be $\mathrm{WN}(0, \sigma^2)$. Let

$$X_t = \epsilon_t + \epsilon_{t-1}\epsilon_{t-2}.$$

   Determine if $\{X_t,\ -\infty < t < \infty\}$ is a white noise process and/or a MDS.